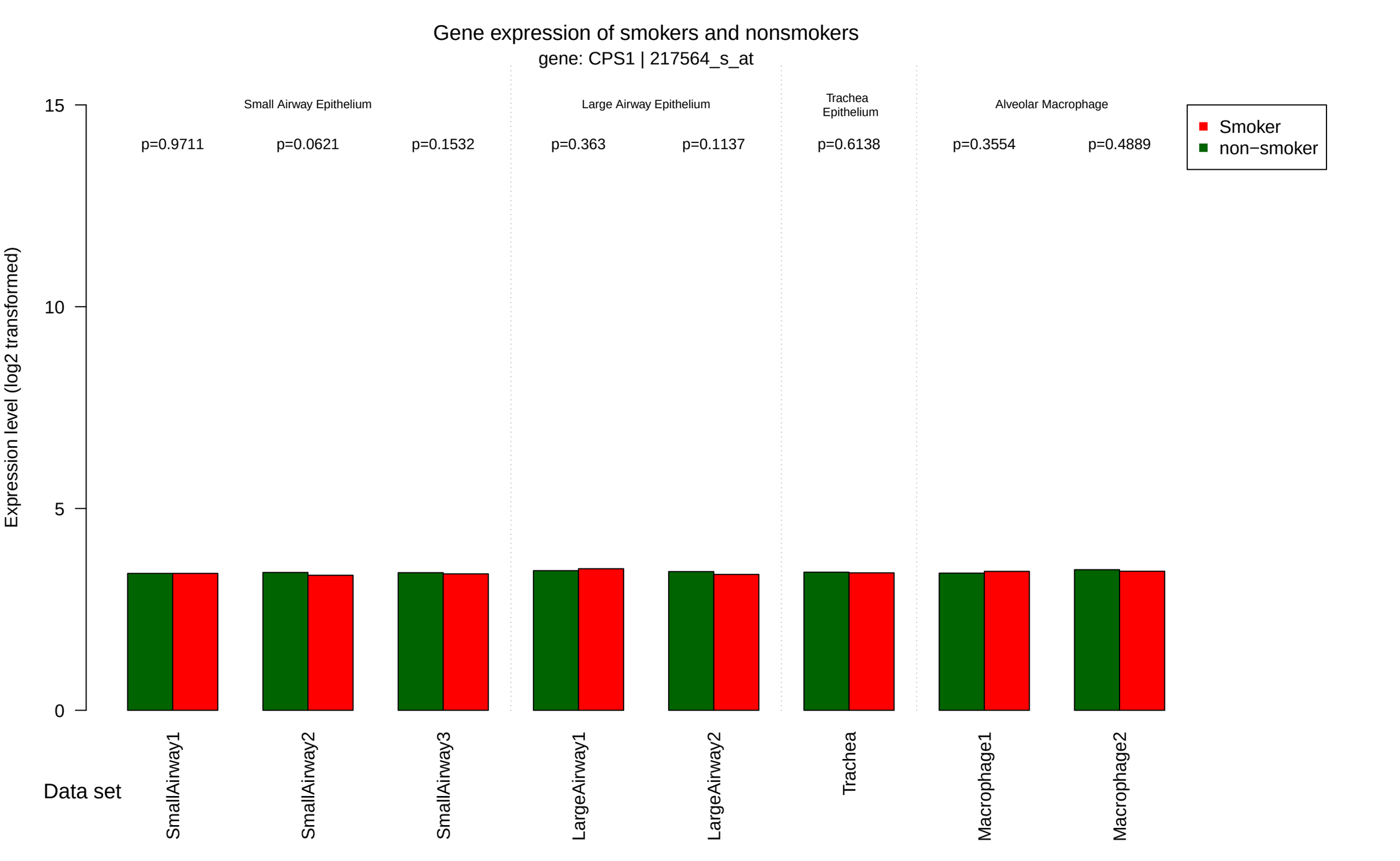
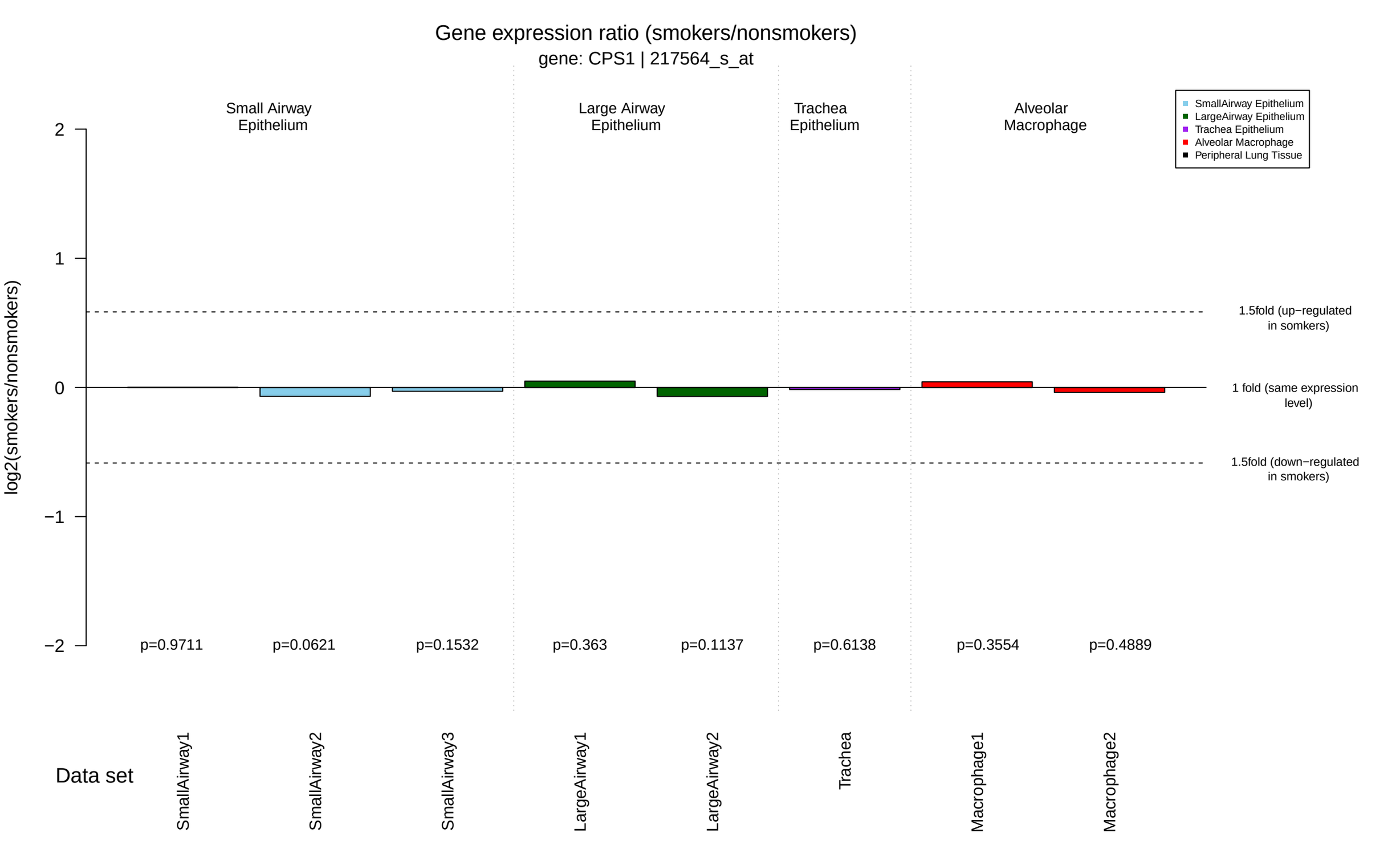
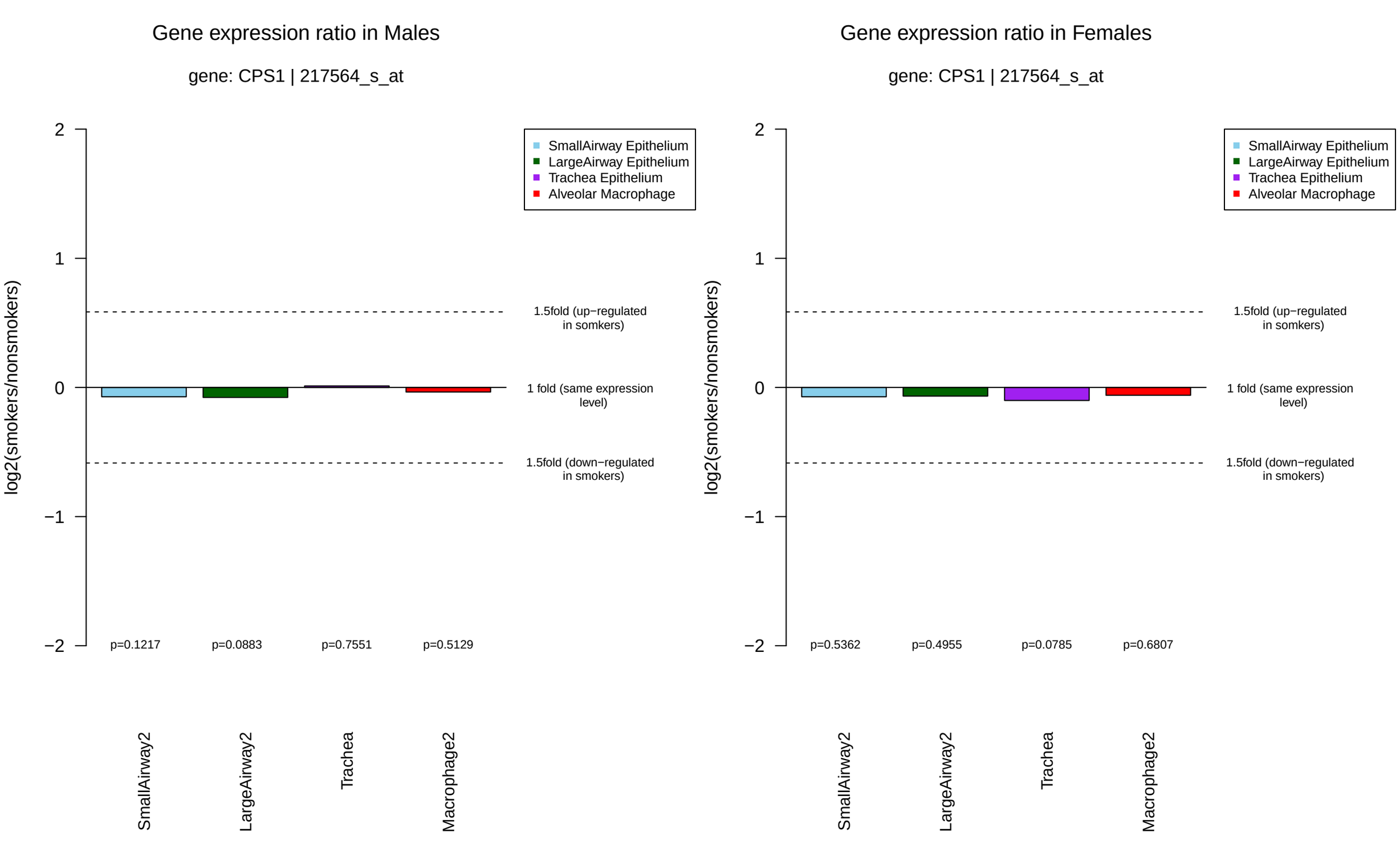
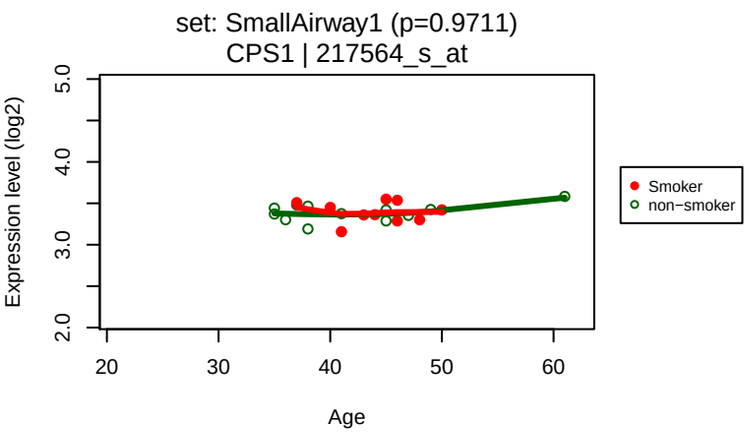
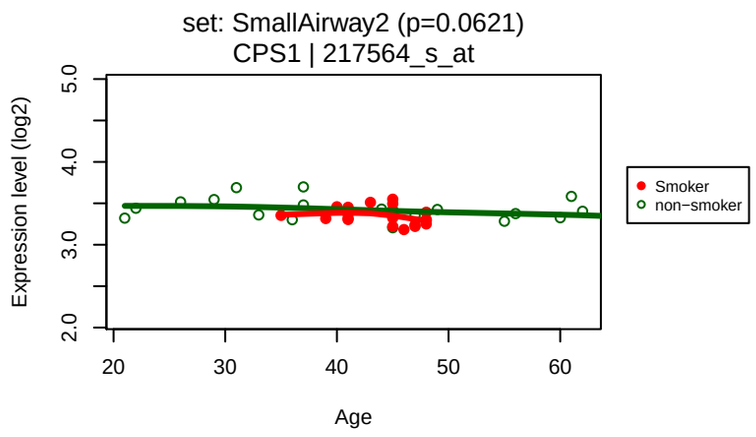
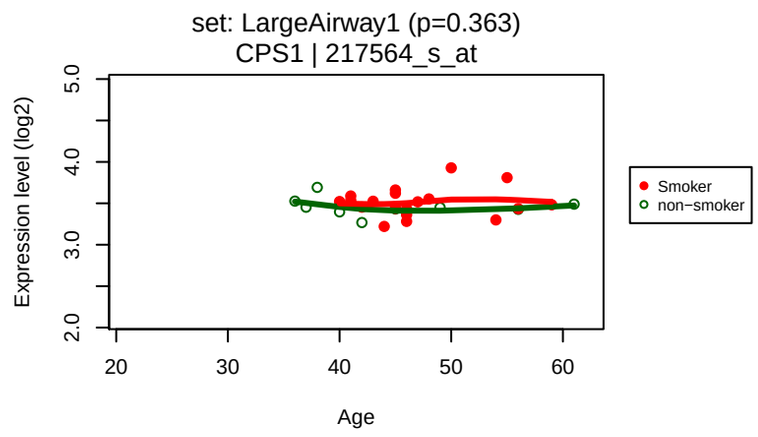

Gene expression of smokers and nonsmokers
gene: CPS1 | 217564_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9711
p=0.0621
p=0.1532
p=0.363
p=0.1137
p=0.6138
p=0.3554
p=0.4889
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

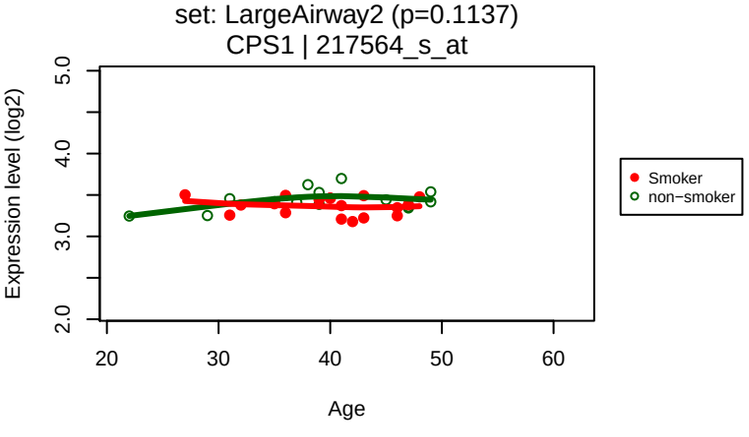

Gene expression ratio (smokers/nonsmokers)
gene: CPS1 | 217564_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9711
p=0.0621
p=0.1532
p=0.363
p=0.1137
p=0.6138
p=0.3554
p=0.4889
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

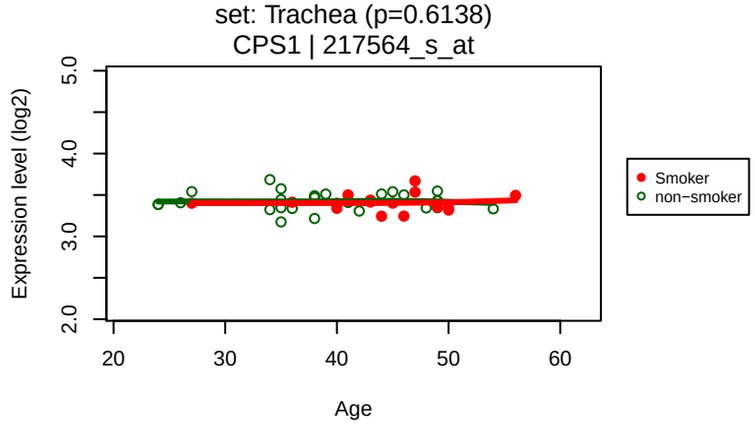

Gene expression ratio in Males
gene: CPS1 | 217564_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1217
p=0.0883
p=0.7551
p=0.5129
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CPS1 | 217564_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5362
p=0.4955
p=0.0785
p=0.6807
SmallAirway2
LargeAirway2
Trachea
Macrophage2

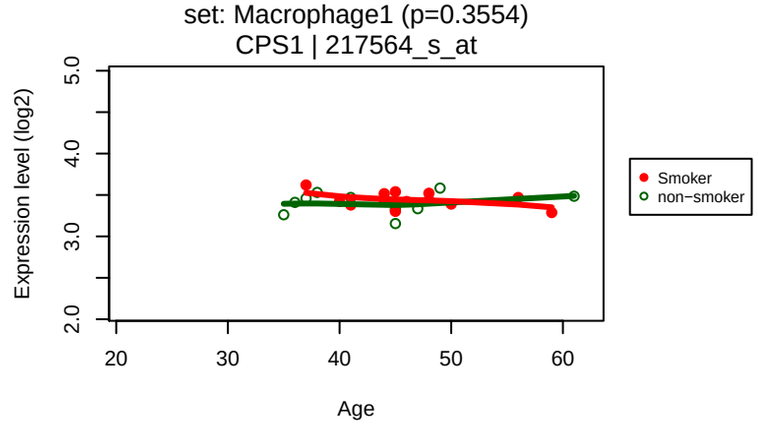

set: SmallAirway1 (p=0.9711)
CPS1 | 217564_s_at
Expression level (log2)
Age
Smoker
non−smoker

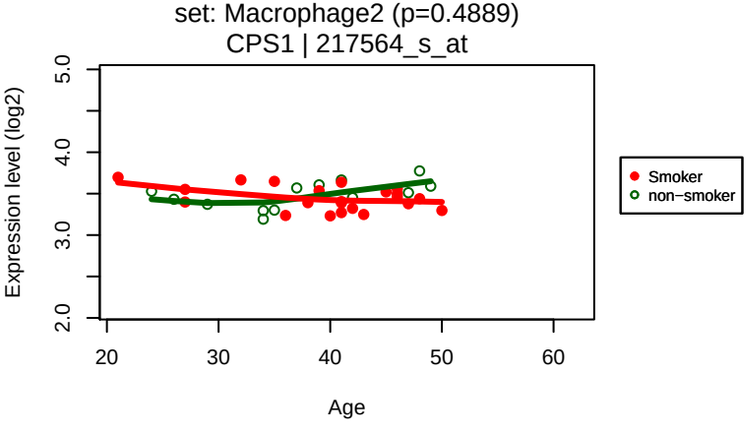

set: SmallAirway2 (p=0.0621)
CPS1 | 217564_s_at
Expression level (log2)
Age
Smoker
non−smoker

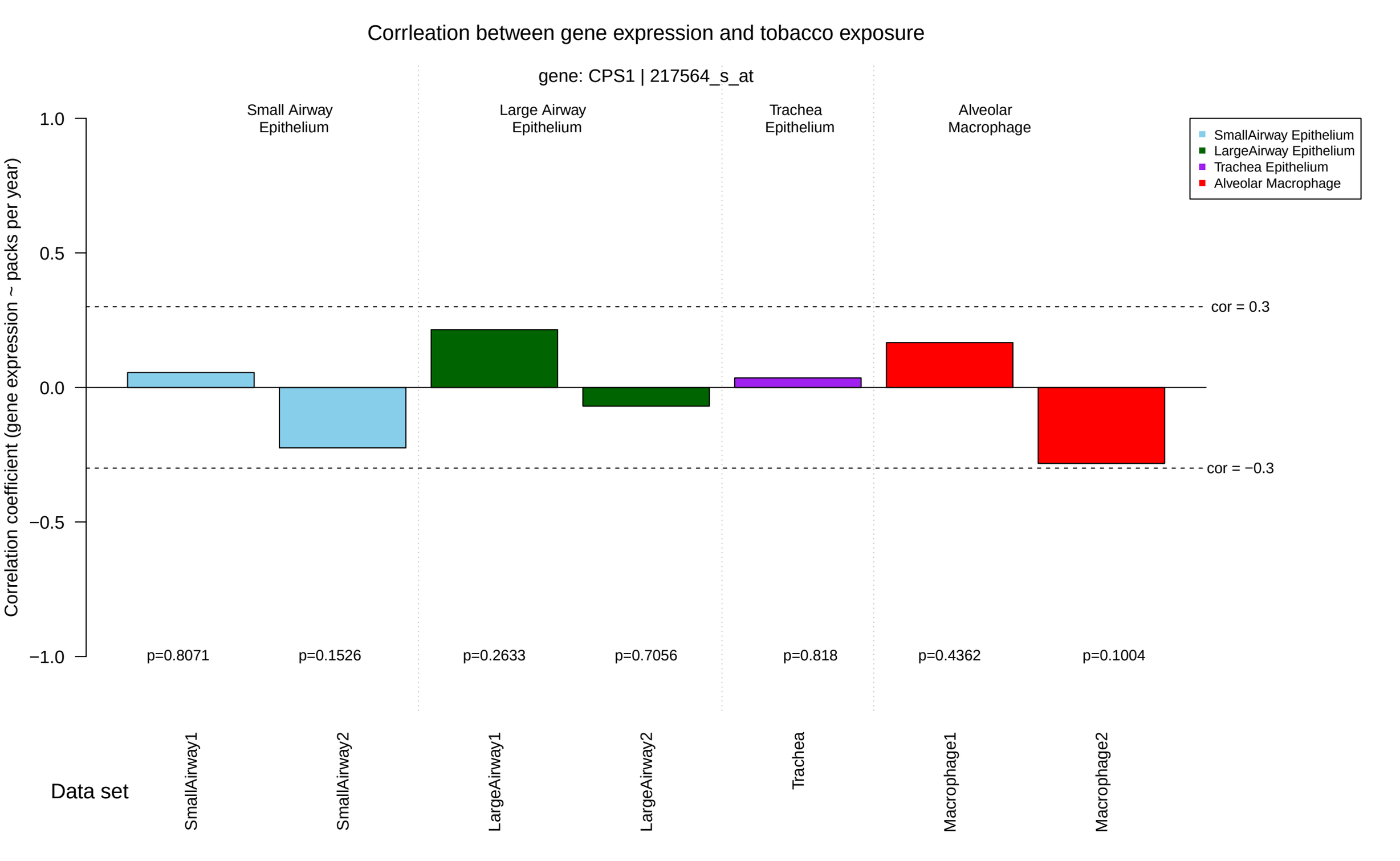

set: LargeAirway1 (p=0.363)
CPS1 | 217564_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1137)
CPS1 | 217564_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6138)
CPS1 | 217564_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3554)
CPS1 | 217564_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4889)
CPS1 | 217564_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CPS1 | 217564_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8071
p=0.1526
p=0.2633
p=0.7056
p=0.818
p=0.4362
p=0.1004
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2